

Metadata of the chapter that will be visualized in SpringerLink

Book Title	Advances in Information Technology in Civil and Building Engineering	
Series Title		
Chapter Title	A Critical Review of Measuring the Modeling Productivity of Building Information Modeling	
Copyright Year	2023	
Copyright HolderName	The Author(s), under exclusive license to Springer Nature Switzerland AG	
Corresponding Author	Family Name	Shin
	Particle	
	Given Name	Sanghyun
	Prefix	
	Suffix	
	Role	
	Division	Building Informatics Group, Department of Architecture and Architectural Engineering
	Organization	Yonsei University
	Address	Yonsei, Republic of Korea
	Email	spicysh03126@yonsei.ac.kr
Author	Family Name	Jang
	Particle	
	Given Name	Suhyung
	Prefix	
	Suffix	
	Role	
	Division	Building Informatics Group, Department of Architecture and Architectural Engineering
	Organization	Yonsei University
	Address	Yonsei, Republic of Korea
	Email	
Author	Family Name	Roh
	Particle	
	Given Name	Hyunsung
	Prefix	
	Suffix	
	Role	
	Division	Building Informatics Group, Department of Architecture and Architectural Engineering
	Organization	Yonsei University
	Address	Yonsei, Republic of Korea
	Email	
Author	Family Name	Lee
	Particle	
	Given Name	Ghang

Prefix
Suffix
Role
Division Building Informatics Group, Department of Architecture and Architectural Engineering
Organization Yonsei University
Address Yonsei, Republic of Korea
Email

Abstract	This study aims to identify indicators to measure the productivity of building information modeling (BIM) through a critical literature review of previous studies on BIM productivity and the factors used in the studies. Measuring BIM productivity, to which end quite a few efforts have been made, is important for efficient workforce management. The authors reviewed 14 papers collated from Scopus and Google Scholar. Examples of topics included productivity factors related to 2D/3D design modeling. Previous studies on BIM productivity have followed the definition of productivity as output/input. The input and output of BIM productivity indicators were classified according to the research purpose of the 14 previous studies. Among the BIM productivity indicators suggested in 14 previous studies, the most frequently used BIM productivity indicators were output as time and input as the number of elements in the 3D model. Among the factors affecting BIM productivity, modeling time was the most commonly considered. Other factors included model quality, modeling behavior, project size, model size and model review.
Keywords (separated by '-')	Building Information modeling - BIM Productivity - Productivity Factor - Productivity Indicator

A Critical Review of Measuring the Modeling Productivity of Building Information Modeling



Sanghyun Shin, Suhyung Jang, Hyunsung Roh, and Ghang Lee

Abstract This study aims to identify indicators to measure the productivity of building information modeling (BIM) through a critical literature review of previous studies on BIM productivity and the factors used in the studies. Measuring BIM productivity, to which end quite a few efforts have been made, is important for efficient workforce management. The authors reviewed 14 papers collated from Scopus and Google Scholar. Examples of topics included productivity factors related to 2D/3D design modeling. Previous studies on BIM productivity have followed the definition of productivity as output/input. The input and output of BIM productivity indicators were classified according to the research purpose of the 14 previous studies. Among the BIM productivity indicators suggested in 14 previous studies, the most frequently used BIM productivity indicators were output as time and input as the number of elements in the 3D model. Among the factors affecting BIM productivity, modeling time was the most commonly considered. Other factors included model quality, modeling behavior, project size, model size and model review.

Keywords Building Information modeling · BIM Productivity · Productivity Factor · Productivity Indicator

1 Introduction

During a building information modeling (BIM) project, efficient manpower management can be achieved by measuring the modeling productivity of each modeler [1]. Through the productivity analysis of each modeler, the project manager can carry out an assigned task flexibly and improve the efficiency of the project team [2]. Modeling productivity can be evaluated by analyzing the working pattern data of each modeler's

S. Shin (✉) · S. Jang · H. Roh · G. Lee
Building Informatics Group, Department of Architecture and Architectural Engineering, Yonsei University, Yonsei, Republic of Korea
e-mail: spicysh03126@yonsei.ac.kr

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2023
S. Skatulla and H. Beushausen (eds.), *Advances in Information Technology in Civil and Building Engineering*, Lecture Notes in Civil Engineering 357,
https://doi.org/10.1007/978-3-031-35399-4_33

BIM model. Identifying individuals' modeling productivity makes it possible to estimate their performance in a specific period or on a project type to manage the project manhour efficiently [1]. Modeling workers can also self-diagnose their shortcomings through productivity analysis and improve their modeling productivity by sharing the command patterns of highly productive workers [3].

Productivity is the ratio of output per input [4]. This paper refers to an index used to represent productivity as "productivity indicator" and inputs and outputs of productivity as "factors".

In BIM productivity, no dominant indicator is used as the output of BIM productivity. Accordingly, determining how to select and measure factors that affect BIM productivity, such as quality, accuracy, and quantity of output, remains a difficult task [5]. BIM productivity indicators and factors are used in various ways, depending on the research purpose. The indicator of BIM productivity is still complicated and difficult due to the complexity of BIM projects because of the absence of standardized and integrated productivity indicators [6]. Therefore, it is necessary to identify and consider BIM productivity factors along with BIM productivity indicators. This study focuses on identifying and classifying BIM productivity indicators and factors. This study has two specific research questions:

1. What "BIM productivity indicators" were used in previous studies?
2. What "factors" were considered as inputs and outputs for measuring BIM productivity in previous studies?

Previous studies were collected and filtered using a systematic literature review method—the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) protocol [7]. Section 2 describes how the PRISMA protocol was employed in this study. Section 3 reviews the selected literature and identifies BIM productivity indicators and the factors affecting the BIM productivity indicator. Section 4 provides conclusions in response to the research questions.

2 Literature Review Method

The PRISMA protocol, the systematic review method that this study deployed, consists of four stages: identification, screening, eligibility, and inclusion [7]. The following subsections describe how previous studies were collected and filtered according to the four stages.

2.1 Data Identification

To collect research papers that dealt with "the productivity of modeling" or "BIM productivity in construction project", two databases, Scopus and Google Scholar, were selected. A total of 180 papers from 2005 to 2022 were collected through

keyword searches. From Scopus, 155 results were obtained using the keywords “((BIM) OR (3D modeling)) AND (productivity) AND ((measure) OR (assess) OR (evaluate)).” In the case of Google Scholar, 25 results were obtained using “BIM productivity measure” and “measurement of BIM productivity” as keywords. Through this process, researchers’ interest in BIM productivity is found to have increased steadily since 2005 and risen sharply since 2013. It is shown that research on BIM productivity analysis has been going on for more than 15 years, and an increasing number of researchers are becoming interested.

2.2 Data Screening, Eligibility, and Inclusion

In the screening step, 128 papers were excluded through a manual review of the titles and abstracts. Four of the excluded cases were papers collected in duplicate from the Scopus and Google Scholar databases. The other 124 cases were excluded for the following reasons: 1) irrelevance with construction, and 2) systematic review papers provided a small amount of insight into the method of measuring BIM productivity.

In the eligibility step, all 52 papers were thoroughly reviewed and excluded from the analysis for the following reasons. Studies on the measurement of worker productivity at construction sites and on the measurement of productivity throughout the construction project process were excluded because they were not related to the modeling productivity measurement method. Moreover, studies on 2D design productivity measurement were excluded because this paper mainly focuses on the productivity measurement of BIM and 3D modeling. As a result, 36 papers were excluded, and 14 were selected. The final 14 papers were reviewed in detail.

3 Analysis Results

The analysis results are reported in two subsections, each corresponding to each research question. Section 3.1 identifies the “productivity indicator” used for BIM productivity measurement and classifies the indicators according to the research purpose. Section 3.2 identifies the input and output “factors” used in the BIM productivity indicator.

To determine the trends of the 14 selected papers, the publication year was analyzed. Two papers were published each in 2005, 2018, and 2020, and one paper each in other years. This shows that research on BIM productivity does not increase but was steadily conducted from 2005 to 2021. Among the 14 selected papers, 3 (21%) are from conferences and 11 are from journals (79%). In terms of publication venue, three papers were published in “Automation in Construction”, three papers in “Journal of Construction Engineering and Management”, and two papers in “Applied Sciences”.

3.1 Productivity Indicators Used for BIM Productivity Measurement

Table 1 summarizes the outputs and inputs of the BIM productivity indicators collected from 14 selected studies and classifies them by research purpose. The research purpose of the BIM productivity is divided into benchmark BIM performance, evaluate BIM project modeling productivity, evaluate BIM modeling worker productivity, and compare 2D drafting and 3D modeling productivity.

The outputs and inputs of the BIM productivity indicators are explained below by research purpose.

Evaluate BIM Project Modeling Productivity. Among the 14 studies, 8 studies (56%) measured BIM productivity to evaluate BIM project modeling productivity [2, 11–14, 8–10].

Table 1 Outputs and inputs of BIM productivity indicators of the selected 14 papers

Research purpose	Output of indicator	Input of indicator	Source
Evaluate BIM project modeling productivity	Number of elements in the 3D model	Week	[8, 9]
	Number of element changes	Week/Month	[8, 9]
	Model LoD	Number of coordination meetings	[8, 9]
	Work hours	1000 m ² of modeled floor area	[10]
	Work hours	Number of elements in the 3D model	[2, 11]
	Work hours	Weight (ton) of elements in the 3D model	[12]
	Work hours	Measurement unit	[13]
	Total wall area	Project lead time (minute)	[14]
Evaluate BIM modeling worker productivity	Work days	Number of elements in the 3D model	[1]
	Work seconds	Command pattern	[3]
	Number of commands	Work hours	[15]
Compare 2D drafting and 3D modeling productivity	Work minutes	Number of drawings	[16]
	Work hours	1000 m ² of modeled floor area	[5]
	Work hours	1000 m ² of modeled floor area	[17]

Two studies [8, 9] were related to a cloud-based BIM application called building information modeling cloud score (BIMCS) for automatic data collection and performance benchmarks of BIM workers. BIMCS developed 20 indicators for performance benchmarks and divided them into modeling productivity, effectiveness, model quality, accuracy, usefulness, and economy categories. This paper includes only the three indicators in the modeling productivity category, which is the main focus of this paper. The first indicator of “number of modeling steps/number of elements” shows how many steps a modeler took to develop an element on average. BIMCS suggests that too many steps are inefficient in modeling an element, and, thus, the smaller the modeling steps per element, the higher the modeling productivity. The second indicator of “number of element changes/number of elements” is an indicator showing how many times an element is modified on average during modeling. BIMCS suggests that more corrections are inefficient, and, thus, the smaller the number of changes per element, the higher the modeling productivity. The last indicator of “model LoD/number of coordination meetings” indicates the average number of meetings required to increase LoD one level, and unlike the two indicators, the higher the value, the higher the modeling productivity.

Sacks et al. [10] presented productivity benchmark that can be applied to the implementation of projects in the use of BIM. Modeling productivity was measured through work hours per 1000 m² of modeled floor area for 6 projects from 3 precast companies. Shan et al. [12] measured the BIM project modeling productivity by dividing the work hours by the weight of the elements in the 3D model, and through this, the existing steel connection system and the proposed steel connection system were compared. Leite et al. [11] analyzed BIM productivity by dividing the work hours by the number of elements in the 3D model [11]. Zhang et al. [2] divided the number of elements in the 3D model by the work hours and multiplied it by the estimated budget for normalization. To measure the progress and productivity of the engineering process in more detail, Garcia et al. [13] set different measurement units for disciplines, for example, “linear foot of pipe” for piping, “per tag” for equipment, and “ton” for structural steel, and divided them by the work time. To measure the performance of the wall production line using the case of a residential house with panels, Brown et al. [14] analyzed the total wall area per project lead time (minute).

Evaluate BIM Modeling Worker Productivity. Among the 14 studies, 3 studies (21%) measured BIM productivity to evaluate BIM modeling worker productivity [1, 3, 15]. These studies aimed to measure the modeling productivity of individual modeling workers rather than the BIM project.

Forcael et al. [15] used “number of commands/work hour” as an indicator to analyze the modeler’s behavior and productivity using the journal file, which is a modeling log of Revit, as data. Commands used for modeling were classified into contributory or non-contributory, and the execution ratio of contributory commands and real-time work were analyzed. Also using the BIM log as data, Zhang and Ashuri [3] identified five patterns of the commands used for modeling and the time taken for each command pattern. Yarmohammadi and Castro-Lacouture [1] measured the time taken to model different types of elements, the number of modifications, and the

number of errors through a custom-developed Revit plugin to identify the optimal design team configuration.

Compare 2D Drafting and 3D Modeling Productivity. Among the 14 studies, 3 (21%) measured BIM productivity to compare 2D drafting and 3D modeling productivity [5, 16, 17]. Sacks and Barak [5, 17] compared the productivity of 2D drafting and 3D modeling in structural engineering design and detailed work of concrete building structures based on the time taken to draft or model 1000 m² of floor area and 1 m³ of concrete, potential benefits, and cost savings in the introduction of BIM. You and Nam [16] analyzed the difference in productivity between 2D drafting and 3D modeling for steel frame prefabrication in a plant facility construction project by measuring the time required to produce one drawing.

4 Productivity Factors Affecting BIM Productivity Indicator

Table 2 shows the factors that affect the BIM productivity indicator of 14 selected studies and data that measured the factors.

Model quality is a factor group that indicates how error-free a model is. It can be judged by the clashes and errors occurring within the 3D model. In the case of a clash, it is possible to determine which elements were clashed and whether the clash was positive or negative through automatic clash detection [11]. In the case of an error, analysis can be done based on data from the journal file extracted through an application developed using the Autodesk Revit application programming interface (API). Through this, the number of errors, types of errors, and location of the drawings

Table 2 BIM productivity factors of 14 selected papers

Factor group	Factor	Source
Model quality	Clash	[11]
	Modeling error	[1]
Modeling behavior	Number of modifications of an element	[1, 8, 9]
	Command pattern	[3]
Project size	Floor area	[10, 17]
	Wall area	[14]
	Number of drawings	[16]
	Project estimated budget	[2]
Model size	Number of elements in a BIM model	[2, 8, 9, 11, 13]
	Number of command executions	[15]
Model review	Number of coordination meetings	[8, 9]
Modeling time	Time (month/week/hour/minute/second)	[1–3, 5, 8–11, 13–17]

where the errors occurred can be identified while the modeler is performing BIM modeling [1].

Modeling behavior is a factor group that represents the way in which a modeler authors a model. This factor group includes the number of modifications of an element [1, 8, 9], the number of project data entries [8, 9], the number of command executions [15], and command patterns [3].

Project size is a factor group that represents the overall project size, such as the floor area [10, 17], the wall area [14], number of drawings [16], and the project estimated budget [2]. The project size factors were used mostly as a normalization factor.

The model size factors were used as both an input or output. The largest number of studies that used the number of elements in the 3D model to represent the model size [2, 8, 9, 11, 13]. These studies measured BIM productivity through the work hours it took for a modeler to model elements in a 3D model. The size of the model was likewise judged through the number of commands that the modeler used in the modeling process and used as a factor of productivity [15].

The model-review factor group represents the degree of effort to review a model or design. In this group, only one factor—the number of coordination meetings [8, 9]—was considered in the previous studies.

Modeling time was used as a factor in the analysis of BIM productivity in all 14 studies. This factor refers to the workhours that the modeler worked for modeling. The different units of measurement for time—i.e., second, minute, hour, day, and week—were used in each study, according to the purpose. Among them, hour was the most used unit [2, 5, 10–12, 15, 17]. Other time units, such as second, minute, day, and week, were used relatively less frequently than hour [1, 3, 8, 9, 14, 16].

5 Conclusion

To identify and classify the indicators and factors of BIM productivity, this paper analyzed research papers on BIM productivity using the PRISMA protocol. Papers were collected by keyword search, and an analysis process was conducted focusing on modeling productivity analysis rather than the productivity analysis of an entire BIM project. From the review, it is clear that different indicators and factors are used in each study for each research purpose. The two research questions of this paper can be answered as follows.

Productivity Indicator Used to Measure BIM Productivity

Productivity indicators for measuring BIM productivity can be classified into three groups by research purpose: evaluate BIM project modeling productivity, evaluate BIM modeling worker productivity, and compare 2D drafting and 3D modeling productivity. Among the BIM productivity indicators, the most researched purpose was to evaluate BIM project modeling productivity, and the most used indicator is work time (modeling time) divided by the number of elements in the 3D model.

BIM productivity indicators vary according to the purpose of the study. To evaluate the modeling productivity of the BIM project, studies mainly measure the time spent modeling the elements of the BIM project. Conversely, to evaluate the productivity of each BIM modeler, studies mainly measured the modeling time per the number of used commands or the modeling time per the number of modeled elements. To compare 2D drafting and 3D modeling productivity, studies mainly measure the time taken to draft or model 1000 m² of floor area.

Factors Affecting BIM Productivity

A total of 12 factors were identified as BIM productivity factors from the 14 selected studies: work time (month/week/hour/minute/second) (100%), the number of elements in a BIM model (35%), the number of modifications of an element (21%), the number of coordination meetings (21%), floor area (21%), clash (7%), modeling error (7%), command pattern (7%), wall area (7%), the number of drawings (7%), project estimated budget (7%), and the number of command executions (7%). And these factors can be classified into 6 groups among the 14 studies: modeling time (100%), model size (42%), project size (35%), modeling behavior (28%), model quality (14%) and model review (14%).

The contents to be studied in the future are as follows. This study identified various BIM productivity indicators, but the correlation between them has not been studied yet. Also the identified indicators represent their own productivity value, but does not provide a single integrated BIM productivity value. An integrated BIM productivity measurement model is also required.

Acknowledgements This work is supported by the Korea Agency for Infrastructure Technology Advancement (KAIA) grant funded by the Ministry of Land, Infrastructure and Transport (Grant 22AATD-C163269-02).

References

1. Yarmohammadi S, Castro-Lacouture D (2018) Automated performance measurement for 3D building modeling decisions. *Autom Constr* 93:91–111. <https://doi.org/10.1016/j.autcon.2018.05.011>
2. Hartmann, T., Gao, J., Fischer, M.: An Analytical Model To Evaluate And Compare 3D Modeling Productivity On Construction Projects, (2006)
3. Zhang L, Wen M, Ashuri B (2018) BIM Log Mining: Measuring Design Productivity. *J Comput Civ Eng* 32:04017071. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000721](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000721)
4. AlChaar E, Issa CA (2020) Engineering Productivity Measurement: A Novel Approach. *J Constr Eng Manag* 146:04020087. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001875](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001875)
5. Sacks R, Barak R (2008) Impact of three-dimensional parametric modeling of buildings on productivity in structural engineering practice. *Autom Constr* 17:439–449. <https://doi.org/10.1016/j.autcon.2007.08.003>
6. Wong JH, Rashidi A, Arashpour M (2020) Evaluating the Impact of Building Information Modeling on the Labor Productivity of Construction Projects in Malaysia. *Buildings* 10:66. <https://doi.org/10.3390/buildings10040066>

7. Moher D, Liberati A, Tetzlaff J, Altman DG (2010) Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *Int J Surg* 8:336–341. <https://doi.org/10.1016/j.ijsu.2010.02.007>
8. Du J, Liu R, Issa RRA (2014) BIM Cloud Score: Benchmarking BIM Performance. *J Constr Eng Manag* 140:04014054. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000891](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000891)
9. Liu R, Du J, Issa RRA, Giel B (2017) BIM Cloud Score: Building Information Model and Modeling Performance Benchmarking. *J Constr Eng Manag* 143:04016109. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001251](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001251)
10. Sacks R, Eastman CM, Lee G, Orndorff D (2005) A target benchmark of the impact of three-dimensional parametric modeling in precast construction. *Journal of the Precast/Prestressed Concrete Institute*. 50:126–139. [https://doi.org/10.1061/\(ASCE\)EI.1943-5541.0000003](https://doi.org/10.1061/(ASCE)EI.1943-5541.0000003)
11. Leite F, Akcamete A, Akinci B, Atasoy G, Kiziltas S (2011) Analysis of modeling effort and impact of different levels of detail in building information models. *Autom Constr* 20:601–609. <https://doi.org/10.1016/j.autcon.2010.11.027>
12. Shan, Y., Goodrum, P., Haas, C., Caldas, C.: Assessing Productivity Improvement of Quick Connection Systems in the Steel Construction Industry Using Building Information Modeling (BIM). 1135–1144 (2012). <https://doi.org/10.1061/9780784412329.114>
13. Garcia G, Golparvar-Fard M, de la Garza JM, Fischer M (2021) Measuring Progress and Productivity in Model-Driven Engineering for Capital Project Delivery. *J Constr Eng Manag* 147:04021009. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001969](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001969)
14. Brown, R., Barkokebas, B., Ritter, C., Al-Hussein, M.: Predicting Performance Indicators Using BIM and Simulation for a Wall Assembly Line. Presented at the July 3 (2019)
15. Forcael E, Martínez-Rocamora A, Sepúlveda-Morales J, García-Alvarado R, Nope-Bernal A, Leighton F (2020) Behavior and Performance of BIM Users in a Collaborative Work Environment. *Appl Sci* 10:2199
16. Yoo M, Ham N (2020) Productivity Analysis of Documentation Based on 3D Model in Plant Facility Construction Project. *Appl Sci* 10:1126. <https://doi.org/10.3390/app10031126>
17. Sacks, R., Barak, R.: A Methodology for Assessment of the Impact of 3D Modeling of Buildings on Structural Engineering Productivity. 1–10 (2012). [https://doi.org/10.1061/40794\(179\)41](https://doi.org/10.1061/40794(179)41)

Author Queries

Chapter 33

Query Refs.	Details Required	Author's response
AQ1	Please confirm if the corresponding author is correctly identified. Amend if necessary.	